

KATHERINE (KATE) MCCARTHY

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Fort Collins, CO

[LinkedIn](#)

SUMMARY

Dedicated **graduate student** with Colorado State University's Ecosystem Science and Sustainability Program. Experience within the environmental sector stems from work as a **sustainability analyst** and through community-based organizations. Currently, I am pursuing a Professional Science Master's with a focus on water resources. I am interested in applying data analysis to **aquatic ecosystem restoration**.

EDUCATION

Professional Science Master's in Ecosystem Science and Sustainability

December 2026

Concentration in Water Resources

GPA: 4.00

Colorado State University, Fort Collins CO

Bachelor of Arts in Environmental Studies

June 2024

Concentration in Geography

GPA: 3.92

LGBTQ+ Studies Minor

University of California, Santa Barbara CA

PROFESSIONAL EXPERIENCE

Isla Vista Trading Post (IVTP), Co-Outreach Director

June 2023 – June 2024

- Coordinated and worked 3-for-Free events, in which clothing donations are given away for free to community members to support sustainable fashion at UC Santa Barbara.
- Created 2nd Annual IVTP Zine by managing the project, facilitating brainstorming sessions, keeping club members on track, proofreading and editing, planning launch parties, and creating funding requests.
- Helped re-established the [IVTP website](#) to increase outreach beyond the local community.
- Community education via presentation to student groups, such as sorority chapters at UCSB.
- Implemented organizational changes to club structure including separate weekly executive board meetings, programming calendar, and leadership guides.

Sustainability Analyst Intern, Voice for Generation Z (VOIZ)

June 2022 – August 2022

- Researched fashion and personal care products to construct 12 sustainability reviews with critical thinking and professional writing.
- Compiled 3 collections of products and ranked them based on sustainability.
- Awarded "Strongest Reviewer" and "Best Collection."

Kappa Kappa Gamma Sustainability Chair

January 2022 – December 2022

- Collaborated with a sustainability-focused UCSB club, Gauchos Go Green (formerly Greeks Go Green), to implement strategies to obtain green certification for the chapter house.
- Educated chapter members on sustainability initiatives, environmental organizations, eco-friendly volunteer opportunities, and personal changes that can improve an individual's sustainability.

PROJECTS

Focused Environmental Impact Report

- Wrote a focused [Environmental Impact Report \(EIR\)](#) for a hypothetical project for combined faculty housing and classroom space on UC Santa Barbara's Campus.
- Compiled on-site research of the proposed project site to determine the existing aesthetic resources, the focus of the EIR.
- Utilized the California Environmental Quality Act, Santa Barbara County Environmental Thresholds and Guidelines Manual, and the UC Santa Barbara Long Range Development Plan to determine the proposed project's impacts to aesthetic resources.

Remote Sensing Vegetation Recovery Model

- Created a remote-sensing model to analyze vegetation recovery in the Yukon-Kuskokwim Delta, Alaska, following a fire in 2015.
- Used NASA APPEARS Application Programming Interface (API) to download point and raster data from region of interest from Sentinel-2 and Landsat 8/9 using the Harmonized Landsat Sentinel satellite data.
- Built and trained an XGBoost remote sensing model using ground-truth above-ground biomass data and point satellite data from 2019.
 - Model built using NDVI, NDMI, and VARI indices.
- Applied the model to rasters from 2014, 2016, 2019, and 2022 to see how above-ground biomass changed over time.
- Presented findings and conclusions of project to peers.

Alaskan Giant Kelp Forest Conservation Plan

- Created a conservation plan to protect [Alaskan Giant Kelp Forests](#), which are classified as Endangered under the IUCN's Red List of Ecosystems.
- Worked collectively with a team to compose executive summary, planning context, planning team, management process, situation analysis, project scope, fundamental objectives, desired outcomes, conservation features, range of strategies, strategy selection, data, knowledge, risks, and uncertainties.
- Utilized Miradi as a project management tool to visualize relationship between elements of the conservation plan.

Isla Vista Trading Post 2nd Annual Zine

- Acted as head of committee to produce [IVTP's 2nd Annual Zine](#), doubling the content from the previous year.
- Created schedule for various zine components to keep committee members on track for project completion.
- Organized funding request for zine and the launch party from Community Affairs Board.
- Compiled, edited, and formatted segments for entire zine.

Personal Website

- Created a [personal website](#) in RStudio using Quarto.
 - Utilized quarto markdown and YAML configuration to create a multi-page website, with a home page, experience page, projects page, and photo gallery.
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